

Chapter 15

Chemical Equilibrium

1. Which one of the following will change the value of an equilibrium constant? A

A) changing temperature

B) adding other substances that do not react with any of the species involved in the equilibrium

C) varying the initial concentrations of reactants

D) varying the initial concentrations of products

E) changing the volume of the reaction vessel

K_c : concentration equilibrium constant.

- 1) The equilibrium-constant expression depends **only on the stoichiometry** of the reaction, not on its mechanism;
- 2) The value of K_c depends only on the particular reaction and **on the temperature**;

2. Which of the following expressions is the correct equilibrium-constant expression for the following reaction? E



A) $\frac{[\text{CH}_3\text{OH}]}{[\text{CO}_2]}$

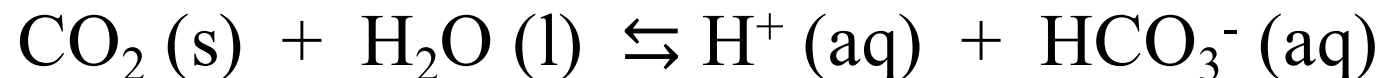
B) $\frac{[\text{CH}_3\text{OH}]}{[\text{CO}_2][\text{H}_2]}$

C) $\frac{[\text{CO}_2][\text{H}_2]^2}{[\text{CH}_3\text{OH}]}$

D) $\frac{[\text{CO}_2][\text{H}_2]}{[\text{CH}_3\text{OH}]}$

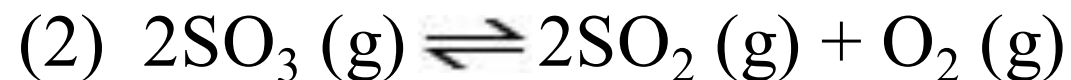
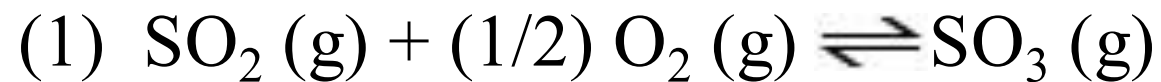
E) $\frac{[\text{CH}_3\text{OH}]}{[\text{CO}_2][\text{H}_2]^2}$

3. Which of the following expressions is the correct equilibrium-constant expression for the reaction below? E



- A) $[\text{H}^+][\text{HCO}_3^-] / [\text{CO}_2]$
- B) $[\text{CO}_2] / [\text{H}^+][\text{HCO}_3^-]$
- C) $[\text{H}^+][\text{HCO}_3^-] / [\text{CO}_2][\text{H}_2\text{O}]$
- D) $[\text{CO}_2][\text{H}_2\text{O}] / [\text{H}^+][\text{HCO}_3^-]$
- E) $[\text{H}^+][\text{HCO}_3^-]$

4. The equilibrium constant for reaction (1) is K . The equilibrium constant for reaction (2) is D.



A) K^2

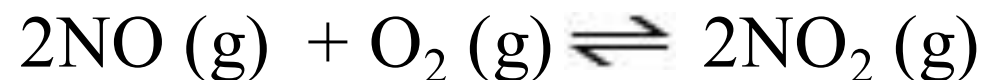
B) $2K$

C) $1/2K$

D) $1/K^2$

E) $-K^2$

5. Given the following reaction at equilibrium, if $K_c = 5.84 \times 10^5$ at 230.0°C , $K_p =$ B.



A) 3.67×10^{-2}

B) 1.41×10^4

C) 6.44×10^5

D) 2.40×10^6

E) 2.41×10^7

$$K_p = K_c(RT)^{\Delta n}$$

$$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$$

6. The equilibrium constant for the gas phase reaction

$\text{N}_2 (\text{g}) + 3\text{H}_2 (\text{g}) \rightleftharpoons 2\text{NH}_3 (\text{g})$ is $K_{\text{eq}} = 4.34 \times 10^{-3}$ at 300°C . At equilibrium, **B**.

A) products predominate

B) reactants predominate

C) roughly equal amounts of products and reactants are present

D) only products are present

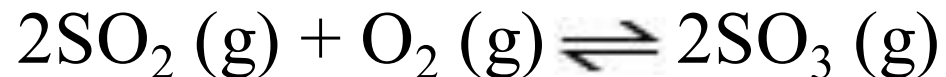
E) only reactants are present

$K_{\text{eq}} = \frac{(\text{moles of gaseous products})}{(\text{moles of gaseous reactants})}$

If $K \gg 1$ (large K): equilibrium lies to right, products predominate;

If $K \ll 1$ (small K): equilibrium lies to left, reactants predominate;

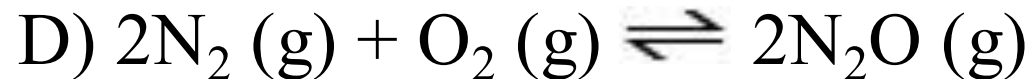
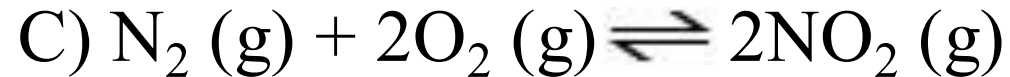
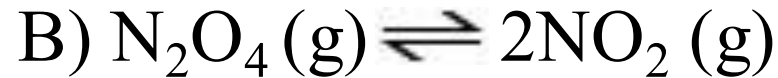
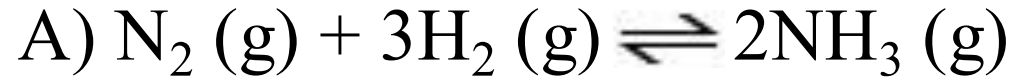
7. Consider the following equilibrium.



The equilibrium cannot be established when B is/are placed in a 1.0-L container.

- A) 0.25 mol SO_2 (g) and 0.25 mol O_2 (g)
- B) 0.75 mol SO_2 (g)
- C) 0.25 mol of SO_2 (g) and 0.25 mol of SO_3 (g)
- D) 0.50 mol O_2 (g) and 0.50 mol SO_3 (g)
- E) 1.0 mol SO_3 (g)

8. In which of the following reactions would increasing pressure at constant temperature change the concentrations of reactants and products, based on Le Châteliers principle? E



E) all of the above

9. The K_p for the reaction below is 1.49×10^8 at 100.0°C : $\text{CO (g)} + \text{Cl}_2 \text{ (g)} \rightarrow \text{COCl}_2 \text{ (g)}$. In an equilibrium mixture of the three gases, $P_{\text{CO}} = P_{\text{Cl}_2} = 1.00 \times 10^{-4} \text{ atm}$. The partial pressure of the product, phosgene (COCl_2), is A atm.

- A) 1.49
- B) 1.49×10^{16}
- C) 6.71×10^{-17}
- D) 1.49×10^4
- E) 1.49×10^{12}

10. Consider the following chemical reaction:



In an experiment, 0.42 mol of CO and 0.42 mol of H₂ were placed in a 1.00-L reaction vessel. At equilibrium, there were 0.29 mol of CO remaining. What is the value of equilibrium constant K_c at the temperature of the experiment?

Answer: 17.5